Sanbot ELF

User Manual

(Multilingual Version)

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Preface

- Thank you very much for purchasing our products. This manual is applicable to the multilingual version of Sanbot ELF (version number is v1.10.41.118).
- Main functions: Sound source localization, live chat, audio-visual entertainment, dance display, greeter, safety protection, auto charging, mobile phone remote control, etc.
- Applicable people: General purpose.
- Suitable age: 12+ years old. Adults need to guide children aged 3-12.
- Please teach children not to touch the joints between the robot's head and body and between the body and wing.
- Our company will regularly improve and update the manual. Updated contents will add into the new edition of this manual without further notice.
- This manual may have some inaccurate technical contents or operational contents not conforming to the product. If you confront with problems that you cannot handle according to this manual, please contact our customer service staff.
- Contents of this manual are only for reference. Please prevail in kind.

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1. To Know Your Robot

1.1 Packing List

Robot

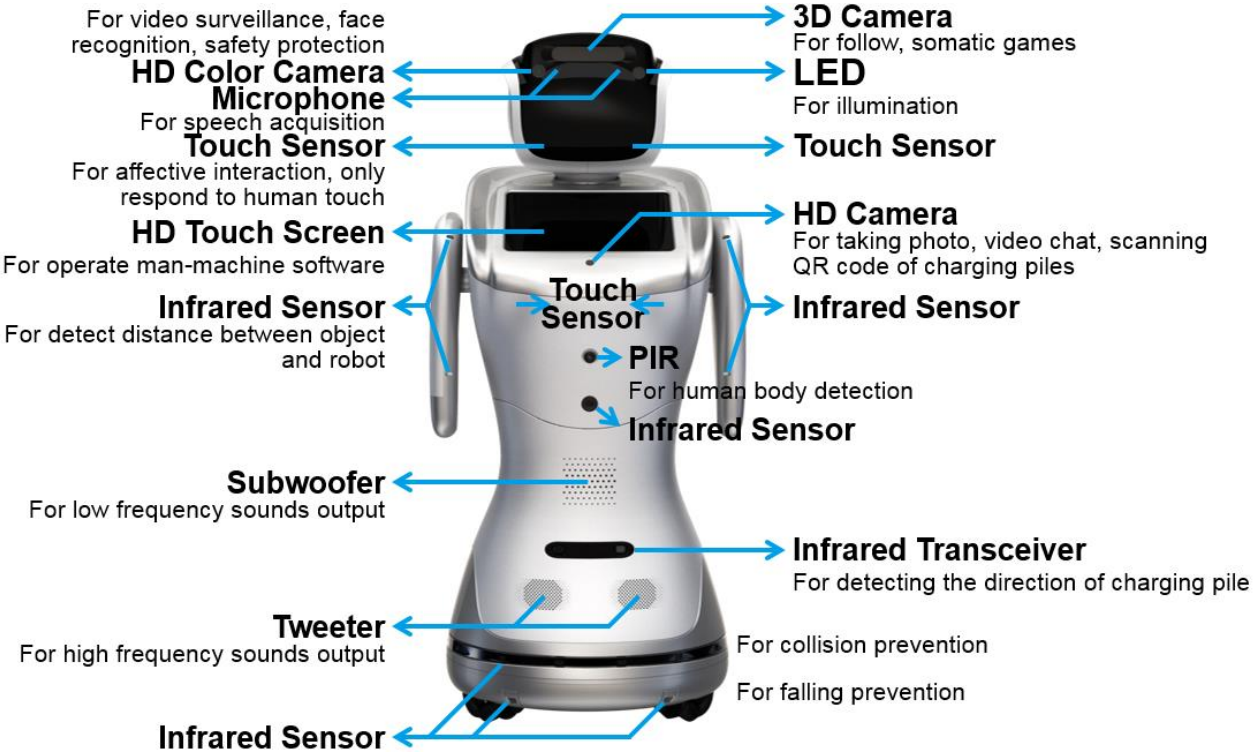
Charging pile (contains adaptor)

Power line

Quick guide manual

We will arrange the consignment according to your choice of purchase; the above picture is for reference only.

1.2 Robot Appearance



Front

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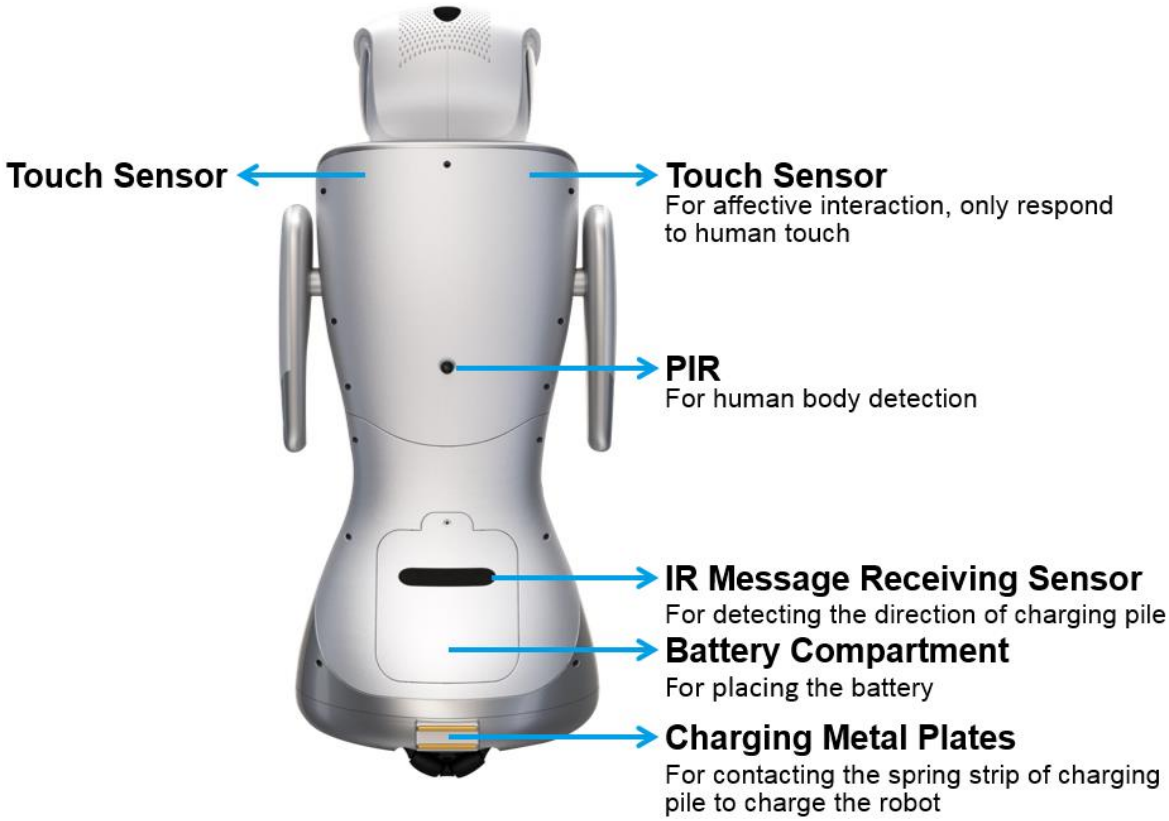
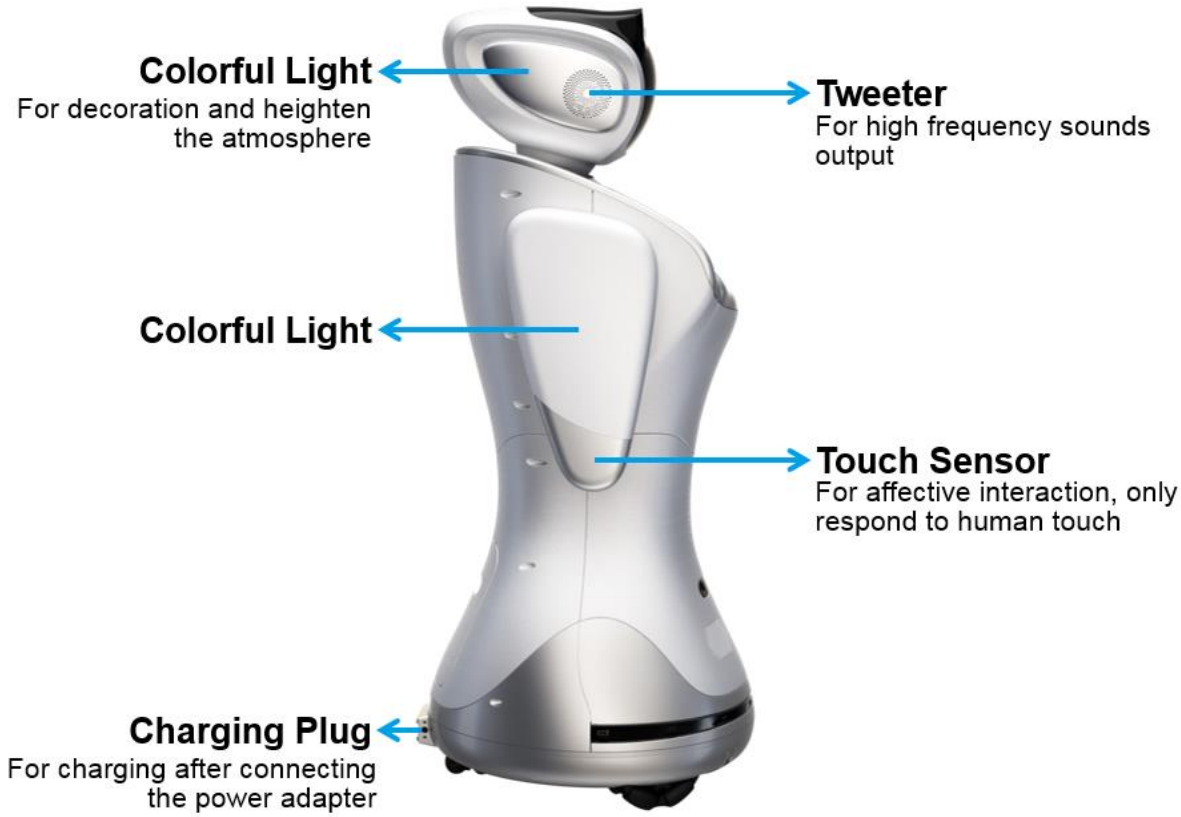


Back



Side

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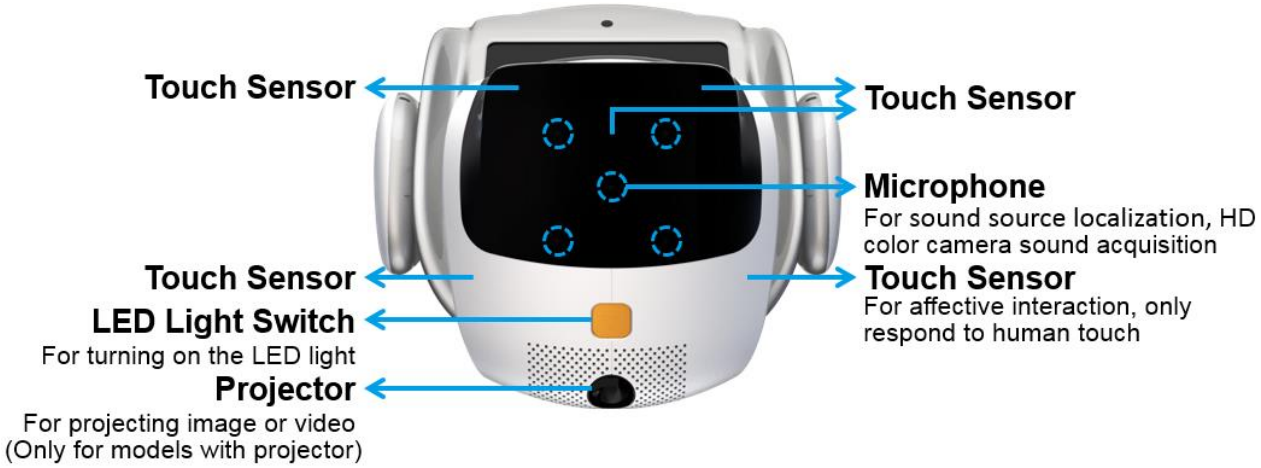
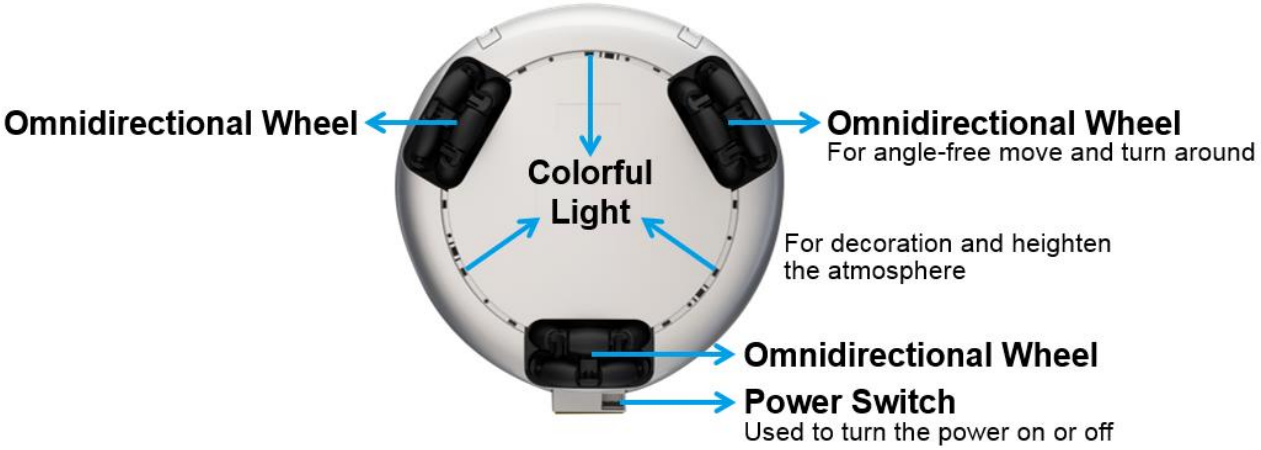


Top



Bottom

1.3 Charging Pile Appearance

Front

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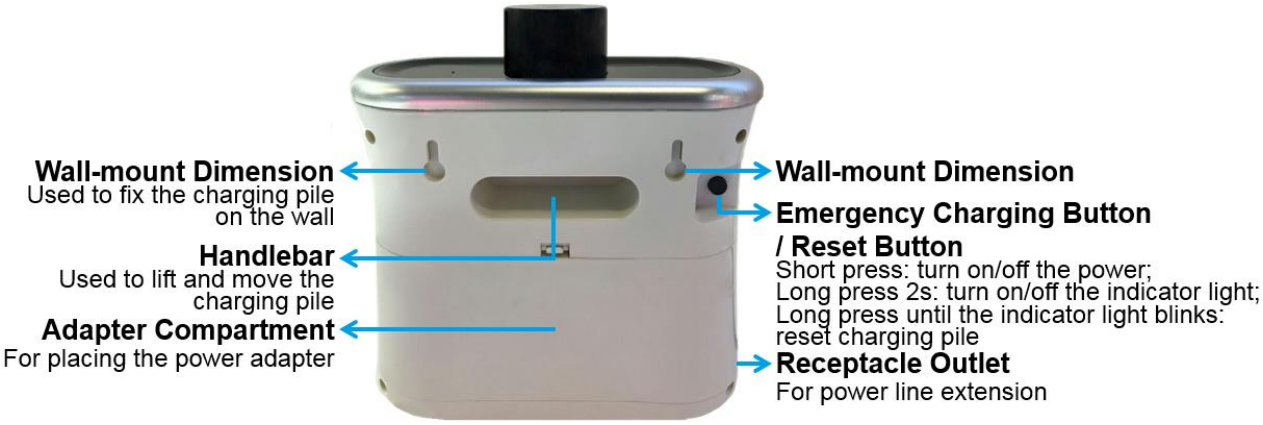

Back

2. Operate the Robot

2.1 Turn On/Off

Press the power switch at the bottom of robot to turn on or turn off the power.

Note: It is easy to move the robot in the state of power-on. However, **when you need to move the robot in the state of power-off, please push its front body forward, and i.e. push it go backwards, or push its back slowly at constant speed.**

2.2 Start-up Guide

After turning on the robot for the first time or restoring the local factory settings, the touch screen will display the wizard:

Boot loading → Select menu language → Agree terms for usage → Select your WIFI and input password → Choose screensaver→ Show initial password (admin) → Scan the QR code by "Q-Link" APP (you can skip this step first) →Click “Finish”. It will not display in the future.

After the robot connects to WIFI, the system needs some time to complete the installation. Please wait a few minutes.

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2.3 Log-in

By default, anyone can use all functions of the robot for convenience. Touch the robot's screen to enter the desktop, click button on the desktop to enter the menu directly.

If you want to verify the password when entering the menu, please open the "Permission" APP, click “ONE KEY ACTIVATE PERMISSIN”, confirm activation and enter the password when the admin logs into the “Q-Link” APP. After that, you need to enter the robot’s login password (default is “**admin**”) when clicking button on the desktop to enter the menu.

Note: It is recommended that you activate the permission and change the password as little as possible; because it will be troublesome for you or easy to forget the password. Please refer to chapter "2.6.7 Permission" for specific operations.

2.4 Desktop Configuration

Touch the robot's screen to enter the desktop, four apps display on the desktop by default. Click these apps on the desktop to open it quickly without entering the menu.

How do I set which apps are displayed on the desktop?

1) Click button on the desktop to enter the menu;

   Note: If you have activated the permission (see chapter "2.6.8 Permission"), you need to enter the robot’s login password (default is “**admin**”) when entering the menu.
2) Open the “Desktop Replace” APP.
3) Select the apps which need to display on the desktop (up to display 4 apps);
4) Click "SAVE".

2.5 Hover Button

Click the hover button to return to the previous menu.

Long press and drag button to change its position.

Long press button to pop up an adjustment window. The function of buttons in the window are as follows:

1) Click the control bar of or drag the slider to adjust the volume;
2) Click the control bar of or drag the slider to adjust the brightness;
3) Click button to turn on/off WIFI;
4) Click button to enter “Message Center” APP;
5) Click button to return to the robot's desktop;
6) Click button to prevent the robot from being woken up by the ambient sound, and the ear light turns blue. Click it again to cancel the disabled state;

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7) Click button to take a screenshot;

If there is no operation about 5 seconds, this window will close automatically.

2.6 Menu Operation

The robot will return to the screensaver automatically in some interfaces if it does not perform any operation in about 24 seconds (can be modified in "Settings") without being woken up. Applications that require the screen to light all the time do not return to the desktop, such as movie, music, dance, and some third-party apps.

2.6.1 Settings

Wi-Fi: Enable the wireless network of robot, select your WIFI and input the password to connect. Click “Other” to connect the WIFI whose name is hidden. After connecting to the Internet, it will display the IP address and MAC address.

Bluetooth: Click the device name to send a Bluetooth pairing request after enabling Bluetooth. Click "More operations" to make "Rename Tablet", "Visibility Timeout" and "Show files received" settings. Click "Search for device" to refresh the device list. Click the robot name to choose whether to allow the nearby Bluetooth devices or paired devices detect this robot.

Cast: You can project the robot's screen content to a large screen through a wireless display adapter. Please refer to chapter "5.2 FAQ" for specific operations. After enabling "Projected voice", the robot's voice no longer play by the robot, but by the large screen.

Display & Brightness: Click the control bar or drag the slider to adjust the brightness.

Voice: Click the control bar or drag the slider to adjust the volume (music, movie, games, alarm, etc.).

Robot: The function of each option refer to the following table:

Face selection	Change the screensaver.
Voice wake-up sensitivity	Select the appropriate sensitivity according to the use environment, user personality and other factors. Higher sensitivity makes it easier to wake up the robot, but it also increases the false wake-up rate.
Wait time for screen saver	After the specified period with no operation, the system will return to the screensaver automatically.
Black line filtration	When this function is not enabled, the robot will stop moving when it encounters wide black bars and dark carpets on the ground as obstacles. However, after enabling this function, the anti-fall and obstacle avoidance function will fail, please enable it with caution.
Human detection	After enabling this function, the robot will turn around actively when it detects someone behind it.
Robot take the initiative to say hello	After enabling this function, the robot will detect the face actively and say hello to the person in front of it.
Wake up the dialogue into the interface	When the robot is awakened, it will open the “Zhi Yin” APP automatically.
Automatic app update	After enabling this function, the APP will upgrade to new version automatically when it detects that there is a new version of the APP installed from app market.
Speed, Intonation, Voice volume	Adjust the voice feature by drag the related bar.

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Battery: Open the automatic charging function of robot, and set the power threshold. (Running mode has been cancelled)

LED: Adjust the brightness of robot LED light.

Date & Time: Enable auto NTP, or manually set the robot's system date and time, and select the time format, date format and time zone.

Language & Region: Select the system language. Select the parameters of two other items as required (some sources may not support now, please refer to the actual effect).

About: Display the robot's name, ID, QR code, MAC address, version number, memory information, legal information, etc.

- Click "QR code" and scan it through "WhatsApp" APP (or other app that can scan QR code) to copy the ID of robot.
- Click "QR code" and scan it through "Q-Link" APP to apply for adding this robot. See chapter "3.3.2 Add Robots" for details.
- Click "Factory Reset" and enter the password (default is "**admin**") to restore the robot settings to the factory defaults (Contacts are not included. Admin and other friends will not be deleted).

2.6.2 Zhi Yin

Zhi Yin is an interactive interface of TTS / STT. The interface lets the robot answer your various inquiries, small talk, etc. It displays what it hears and says on the screen. The answer comes from Google's semantic system, which allows robots to have personalized speech expressions. Please read the "Google dialogflow build instruction".

The robot needs to be in a listening state **to hear your question. You can enter the listening state by the following methods.**

1) Click button;
2) Say "Hello Sanbot" to the robot;
3) Touch the sensing position of robot (see Chapter "1.2 Robot Appearance" for details).

You can have a voice conversation with the robot when its ear lights are green. If no voice is detected over a period of time, the system will stop listening automatically.

Click button in the bottom right to lock the dialog window, set the welcome speech (voice content when entering "Zhi Yin" APP) and guide sentences (recommended questions on the "Zhi Yin" interface).

Click button in the upper right to exit the "Zhi Yin" APP. If you have enabled "Password

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verification", you need to input the login password (default is "admin") when exiting this interface.

2.6.3 Contacts

Display all users who have added this robot as a friend. You can select a member to send text messages, voice messages, emoticons, pictures and voice / video chat. When a friend sends a message or voice / video chat to your robot, you can also reply on the touch screen to enjoy your happiness with your family and friends at any time.

You can long press a friend to delete it from the Contacts interface.

Note: The admin is the first person who scan the robot's QR code and click "confirm" on the robot. See chapter "3.3.2 Add Robots" for adding the robot. Each robot can only have one admin. Please operate with caution.

2.6.4 Message Center

Receive all session message, update notification and push information. Click "Message setting" to clear messages, set whether to allow the message reminder, and set the way of reminder. When you have an unread message, the robot's ear lights will flash yellow to remind you.

2.6.5 File Manager

View the robot's local files, including "Picture", "Video", "Music" and "Other Documents". Click button and select the files to copy, cut, delete, etc.

Copy files to the robot:

1) Remove the fixing piece on the battery compartment on the back of the robot, use a screwdriver to take out the screw on the battery cover, and remove the battery cover to see the USB port.

2) Copy the files to U disk, and insert the U disk into the robot's USB port.
3) Open the “File Manager” APP, select "USB", and find the files that need to be copied.
4) Click button and select the corresponding files, click "Copy to" or "Cut to", and then click "Other Documents" to confirm the copy.

It is not recommended that you use this method too much. It takes up the limited capacity of local

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storage, and the robot has an APP Market to enjoy online music, online video, audiobooks, etc.

2.6.6 APP Market

Download or update the APP in the robot at any time.

After downloading and installing the app, it will display in the menu.

Click "Come into my life" and select "Uninstall" to uninstall the apps that are not pre-installed.

2.6.7 Permission

If you want to verify the password when entering the menu, please open the "Permission" APP, click “ONE KEY ACTIVATE PERMISSIN”, confirm activation and enter the password when the admin logs into "Q-Link" APP. After that, you need to enter the robot’s login password when entering the menu and "Permission" APP.

Note: It is recommended that you activate the permission and change the password as little as possible; because it will be troublesome for you or easy to forget the password. The default login password is “**admin**”. The robot must have an admin when using "ONE KEY ACTIVATE PERMISSIN"; otherwise, it will not be available.

- Access restriction: Password authentication is already the design of access restriction. Therefore, this function is cancelled.
- Lock setting: After opening and setting lock function for a certain app, the robot will open this app automatically as you turn on the robot next time. Other apps are hidden unless you unlock them by clicking the hover button and input the admin password (default is “admin”).
- Robot password modification: Modify the robot’s login password (default is “admin”).

2.6.8 Movie

Open the robot's local movie player (see chapter "2.6.5 File Manager" for copying files).

Tap the screen to pop up the control buttons when the video is playing. Support previous, pause \ play, next, play at the specified location, adjust volume, single cycle \ random play \ sequential play, select from the list, etc.

Click "<" button on the right to display the list, and then click to view or delete local videos, click ">" button to collapse the list.

2.6.9 Music

Open the robot's local music player (see chapter "2.6.5 File Manager" for copying files).

When the music is playing, the interface support previous, pause \ play, next, play at the specified location, adjust volume, single cycle \ random play \ play in order, select from the list, etc.

Click "<" button on the right to display the list, and then click to view or delete local music, click ">" button to collapse the list.

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2.6.10 Dance

The robot dancing can gather a large number of customers immediately to watch and take pictures, so it can successfully attract and gather customers for you.

Click button in the bottom right corner of the dance interface to view all saved dances. Click button to edit the dance (the built-in dance cannot be edited); click button to delete the dance. Select a dance to play the dance.

Click button in the bottom left corner of the dance interface to play the current dance in the list. Pause the dance: While the robot is dancing, say "Hello Sanbot" to wake up the robot, or touch the screen and click "PAUSE", or touch the touch-sensitive area (Except the top of head).
Stop the dance: While the robot is dancing, touch the top of head, or touch the screen and click the hover button, etc.

Note: When the robot is dancing, the obstacle avoidance function is turned off. Please ensure that there is enough space around and pay attention to safety. If the robot needs to walk, please exit the dance first.

How to edit a dance?

Method 1: Quick editing.

1) Select a dance mode: Ballet, Waltz, Square and Modern.
2) Select a piece of music and click "Save": the music is robot's local file. See chapter "2.6.5 File Manager" for copying files.
3) Enter the dance name and click "Confirm" to save the dance.

Method 2: Customize editing.

1) Click “Custom Mode”.
2) Select a piece of music and click "Next": the music is robot's local file. See chapter "2.6.5 File Manager" for copying files.
3) Select the dance’s action in the Dance Library. If you think it is too troublesome to design the action yourself, you can also click "One Touch" to select the action randomly. A dance supports a maximum of 16 dance actions.

   Tips: Click "Dance Library" in the dance interface to view and add dance actions. Click "+" to select the movements of four beats for the new action (Select them according to your creativity. It is not necessary to specify the movement for all parts). Click "Save" and input the movement name to add to the Dance Library.
4) Click a dance action twice in the Selected Dance Steps to delete it.
5) Click “Save”.
6) Enter the dance name, select its speed and click "Confirm" to save the dance.

2.6.11 Trumpet

◆ **How to make your robot play voice \ music \ video \ PPT in a loop:**

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1) Click “Create The Content”;
2) Enter the name of trumpet and click "Done";
3) If you need the robot to play a voice (not many words) in a loop, please select "Input text", input the text content, and click "+" to select an action template;
4) If you need the robot to play a longer voice \ a piece of music \ a video \ a PPT in a loop, please select "Select file", select a file and an action template;
   Note: The file (txt\mp3\mp4\PPT) needs to be copied into the robot in advance, see chapter "2.6.5 File Manager" for copying files. It is recommended to use the PPT within 16M. You can set "Play the note of PPT" by voice and page turning time.PPT with animations, hyperlinks, oversized pictures and “.pptx” format is not available.
5) Click "Save";
6) In the "Trumpet Library" list, click a trumpet to view or modify its configuration. Click button to play it in a loop; click button to delete it.

◆ **How to make your robot play the trumpet according to time or other conditions (the trumpet has been created):**

1) Click "Today Task" to switch the page;
2) Click "Add Task" to set the trumpet schedule for today, or select another date and then click "Add Task";
3) Set the start & end time and repeat days of schedule;
4) Click "+" to select the trumpet’s content (up to five), click "↑" or "↓" to adjust the order, click button to delete this content;
5) Select a trigger mode: After selecting "Auto Play", the robot will play the trumpet according to the time and mode you set. After selecting "Trigger Condition", the robot will play the trumpet after it is triggered according to one or more trigger conditions (human body induction, face induction, touch the head, touch the hand) within the time period you set;
6) Select whether to open the projector (only for the robot with projector) : After enabling this item, the robot will open the projector automatically and stop the action of current task when the trumpet is triggered;
7) Click "Save", select a broadcast mode (one time, loop, five minutes loop, ten minutes loop), and then click "Done";
8) In the "Today Task" list, click a task to view or modify its configuration. Click button to delete this scheduled task.

Note: When playing the trumpet, the robot's voice wake-up and touch sensing function are turned off. You need to touch the screen and click the hover button to exit.

2.6.12 Upgrade

View the current version information of robot.

When the robot has a new software version, the robot will prompt you. You can click "Upgrade

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Now" and wait for the upgrade to finish.

If you enable "Auto download with WIFI", the robot will download the new version automatically when it detecting a new upgrade program. The robot will prompt you to install after the download is complete.

2.6.13 Hardware Test

Select the item you want to detect. The robot can detect some items automatically. However, some items require you to follow the prompts to participate. After the detection is completed, the interface will display the results.

2.6.14 Camera

Click camera button or use voice command (see Voice Control) to take photos.

After taking photos, click the photo in the bottom right corner to view or delete.

2.6.15 Wander

After enabling this function, the robot will walk around and avoid people or objects, record the audio & video of environment simultaneously, and save them in the robot. Click "View video" in the upper right corner to view the saved video. Click a video to play back; long press a video to delete.

In the process of wander, you can stop walking by touching the top of head, saying "Hello Sanbot" to wake up, touching the screen and clicking the hover button, or touching the screen and turning off the switch.

2.6.16 Duck Run

This is an interactive game with robot. After enabling this function, the robot can recognize the person standing in front of it, rotate 180° (turn around) and go straight for a distance. If it encounters people or other obstacles again during this process, it will go away (not 180°). The robot always "shyly" avoid the person in front of it, but it will turn back and interact with you again. You can guide the robot to a certain place with this game.

2.6.17 Charge

When a charging pile has been successfully added to the robot (see chapter "2.6.18 My Device" for details) and the power line of charging pile has been plugged into the power socket, if you enable this function, the robot will find the charging pile immediately to charge.

2.6.18 My Device

The charging pile has been added to the robot in factory. Click the name of charging pile on My Device interface to enter the "Charge" APP (see chapter "2.6.17 Charge" for details).

If your robot has not added the charging pile or has been restored the factory settings, you need

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to add a charging pile.

- **Add a charging pile:**
  ① Click “Scan QR code to add devices”;
  ② Scan the QR code on the charging pile;
  ③ Change the device name and click “Confirm”;
  ④ **Reset the charging pile: Long press the button on the back of charging pile until the indicator light of charging pile blinks, release the button right now. If you release the button before or after the indicator light blinks, it will lead to the reset failure.**
  ⑤ Refresh the device list (Click "Refresh" button; Press and pull down the device list, then release it to appear a rotating arrow; Re-enter the My Device interface; Restart the robot). The robot has connected to the charging pile when it displays “Connected”.
- Delete a charging pile: Swipe the device from left to right or right to left, and then click "Confirm".
- Modify the name of charging pile: Click button to modify the device name.
- Ignore the device: If you do not use a device, click “Ignore”. If you want to use it again, click "Add to my device" in "Other devices" to undo the previous action.

2.6.19 Projector (Only for robots with Projector)

- Auto start when play the movie: Set whether to turn on the projector automatically when playing a video.
- Click the switch to turn on or off the projector. You can set the projector parameters after turning on the projector.
- Projection mode: Set to “Wall mode” or “Ceiling mode”. The robot adjusts the vertical angle of head automatically, so that the picture is projected on the wall or ceiling. You can move the robot moderately or adjust the head to adjust the projection position.
- Keystone correction: If the projected picture is not rectangular and has some trapezoidal deformation, you can move the robot, adjust the head rotation angle, or adjust it through "Keystone correction".
- Image settings: If the projected picture is not clear, you can dim the room light, adjust the position of robot, or adjust it through “Image settings”.
- Mirror adjust: You can flip the picture horizontally or vertically through “Mirror adjust”.
- Definition settings: If you can see double or the effect is still not good after using the above methods, you can dim the room light and adjust it through “Definition settings”. Click "+" or "-" according to the screen prompts to make the projected lines coincide.

2.6.20 Reminders

- Assign a timed reminder task to the robot:
  ① Click "New" to add a reminder for today, or select another date and then click "New".
  ② Select a task type: If you select "Voice", enter the content you want to play, click "Finish", and

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click a trumpet icon to listen. If you select "Music", "Dance" or "Trumpet", select a file in the list. If you select “Wander”, the robot will walk around according to the schedule.

③ Set the reminder’s start & end time.

④ Set the reminder’s repeat days.

⑤ Click "Finish".

- View, modify or delete a timed reminder: Click a date to view the intraday reminders on the Reminders interface. Click 🗑 button to delete the reminder. Click a reminder to modify it.
- Delete reminders in batches: Click the button in the upper right corner to delete reminders in batches.

2.6.21 Photos

View the robot’s photos or other pictures.

Click "Select" in the upper right corner, select the file to share the picture (send it to the "Q-Link" APP of robot friend) or delete the picture.

Tips: If the interface displays "Send message failed" after sharing the picture, open the “Contacts” APP, select the member to enter the chat interface, long press the shared picture and click "Retry". In this way, his / her "Q-Link" APP can receive your shared picture.

2.6.22 Greeter

The robot can display promotional images or videos, sense the arrival of customers automatically, and take the initiative to greet them.

Step 1: Prepare files.

Skip this step if you want to display the robot's default animation when welcoming guests.

If you want to display your own images or videos when welcoming guests, please configure the files according to the following requirements.

① Create a folder named "usher" on your PC;

② Create 2 subfolders named "images" and "videos" in the "usher" folder;

③ If you want to display images when welcoming guests:

Put the welcome images in the "images" folder. **The best resolution of image is 1920*1200. Support jpg or png format. The name format is "usher_number"**, such as usher_1, usher_12, usher_23, etc.

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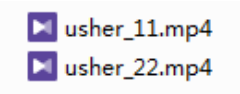
④ If you want to play videos when welcoming guests:

Put the welcome videos in the "videos" folder. **The best resolution of video is 1920*1200. Support mp4 or 3gp format. The name format is "usher_number"**, such as usher_1, usher_12, usher_23, etc.



⑤ Copy the "usher" folder to the **root directory** of U disk, and unplug the U disk after the copy is complete.

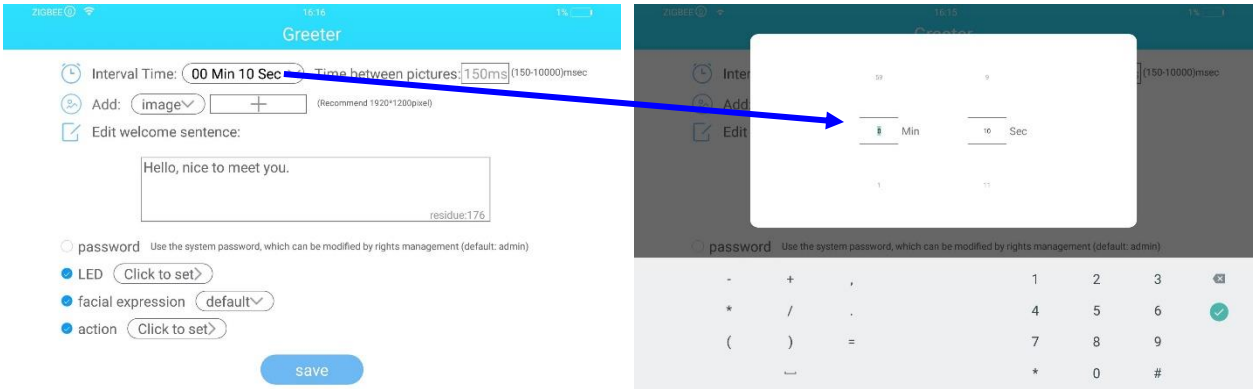
Tips: The files used for greeter must meet the requirements. The "usher" folder must contain two subfolders, and their names are inalterable. It is not necessary to put files in each subfolder, but the files must put in the corresponding subfolder. The format and name must meet the requirement. Otherwise, the "Greeter" APP will not recognize these files.

Step 2: Set up.

① Insert the U disk into the USB port of robot, open the “File Manager” APP, select "USB", click button and select "usher" folder, click "Copy to", and then click "Other Documents".

② When the copy is complete, open the "Greeter" APP and click "Setting".

③ Set the interval time (refer to the time interval for restarting detecting people after detecting a person).



Tips: If you want to display the robot's default animation when welcoming guests, you don't need to

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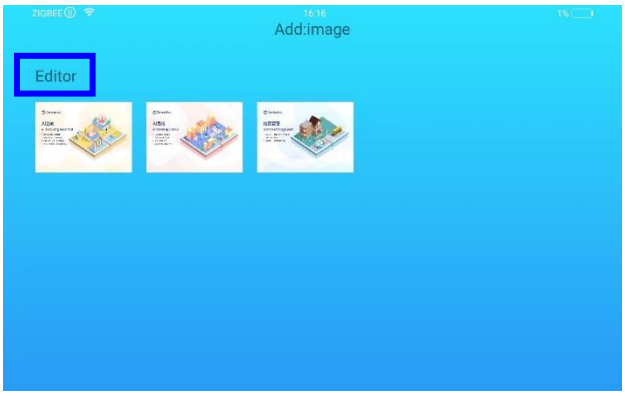
select the file type, skip the content of ④, ⑤.

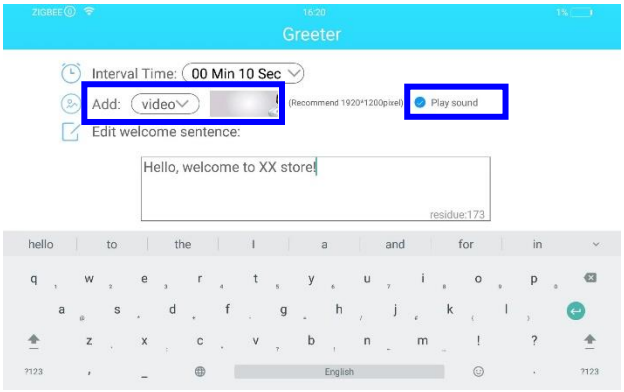
④ If you want to display images when welcoming guests:

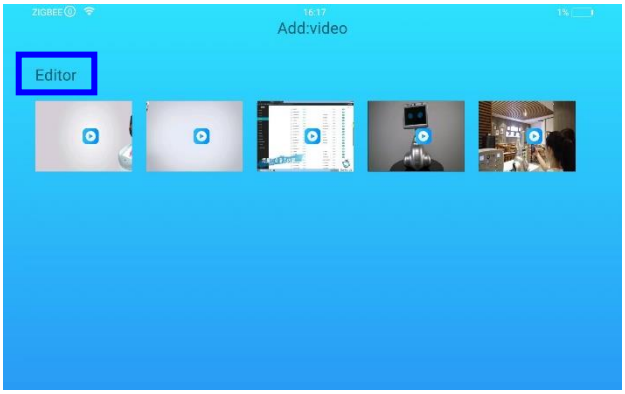
Select "image" type and click "Editor" to select images, and then click "Confirm". Click "+" or picture thumbnail to change the images. Set "Time between pictures" as needed, the unit is “ms”.



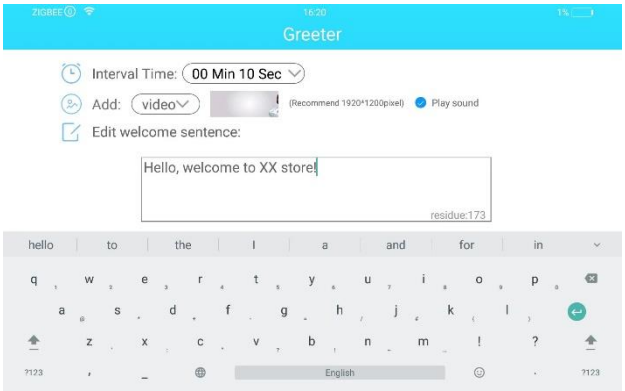
⑤ If you want to play videos when welcoming guests:

Select "video" type and click "Editor" to select videos, and then click "Confirm". Click "+" or video preview to change the videos. After checking "Play sound", the robot can play sounds in the video and say welcome sentence when welcoming guests. Otherwise, the robot just say welcome sentence and play video (no sound) when welcoming guests.





⑥ Edit the welcome sentence (not too long), and then click the hover button to put away the soft keyboard.



⑦ If you check "password", you need to input the login password of robot (default is "admin") when

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exiting "Greeter" APP. If you uncheck it, you can exit APP without a password.

⑧ If you check “LED”, you need to select the light effect when welcoming guests.

⑨ If you check “facial expression”, you need to select the eye expression when welcoming guests.

⑩ If you check “action”, you need to select the action of head, left hand and right hand when welcoming guests.

⑪ Click "save".

Step 3: Start to work.

① Click "OPEN" to enter the greeter mode: The robot can display images or videos, and sense the arrival of customers automatically. When a person is detected, the robot says the welcome sentence automatically and detects again after reaching the interval time.

② When the robot is in greeter mode, tap the screen to exit. If you check "password" in the setting, you need to input the password (default is "admin") to exit.

Tips: After opening the greeter mode, the robot does not respond or respond after a long time if someone passes in front of the robot. Please refer to the following operations: (1) Shorten the interval time. (2) Open the "Hardware Test" APP, select IR Sensor, HD Camera and other related items to judge whether the hardware is faulty.

2.6.23 Marketing Publicity

The operation is similar to the "Trumpet" APP. Here you can create marketing publicity content with text, music or video.

By the way, you can select template (Including Free Walking, etc.) for robot action. Robot will play your marketing publicity in a loop when you open the specific marketing publicity. If the marketing publicity content is text or music, the marketing publicity screen will display the default theme.

2.6.24 Product Introduction

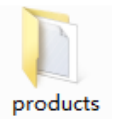
Robot can introduce your products to customers and become your free propagandist.

2.6.24.1 Import Files

It will prompt "No configuration file was imported, please import the configuration file!" if you do not import the content. Please import the product introduction content to the robot.

Step 1: Follow the requirements below to configure the files in your PC. (You can also contact our staff to get the file template, and then modify it)

① Create a folder named "products" on your PC;



② Create two subfolders named "files" and "rules" in the "products" folder, then create a text file

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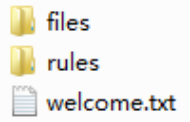
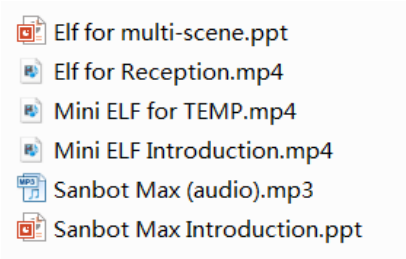
named "welcome.txt" (support UTF-8 and GBK encoding);



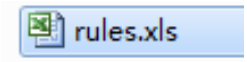
③ In the "files" folder, put the introduction files. Support picture, video, audio, ppt:



Tips:

- Supported video formats: mp4, 3gp, avi, rmvb, flv, mkv, mov, mpg, etc. The maximum size of video file is 600M.
- Supported audio formats: mp3, wma, wav, amr, aac, ac3, etc. The maximum size of audio file is 100M.
- Supported image formats: jpeg, jpg, png, bmp. The best resolution is 1920*1200. It is not recommended that you put some images here. You can put the images into PPT (support to read out the remarks) or "Greeter" APP.
- Supported PPT format: ppt. The maximum size of PPT file is 100M. It is recommended to use PPT within 10M. PPT with animations, hyperlinks, oversized pictures and ".pptx" format is not available here. Please convert to ".ppt" format of PowerPoint 97-2003.

④ Create an Excel file named "rules.xls" in the "rules" folder;



Tips: ".xlsx" format is not available. Please convert to ".xls" format of Excel 97-2003.

⑤ Open "rules.xls" and fill in your product introduction information according to the following instruction and diagram;

- filename: Be consistent with the filename in the "files" folder.
- number1: Set the classification of each file and the order of classification in the navigation bar. Same category's "number1" is equal.
- number2: Set the order of each file. "number2" are not repeatable.
- category: Set the content displayed in the navigation bar for each classification.
- voice1、voice2、voice3: Keywords for speech recognition. You can set up to three keywords for each file (each keyword has a maximum of 16 characters; overlong keyword is not recommended). When a customer say the product keywords they are interested in the product introduction interface, the robot will search for products automatically based on keywords and introduce the corresponding product.

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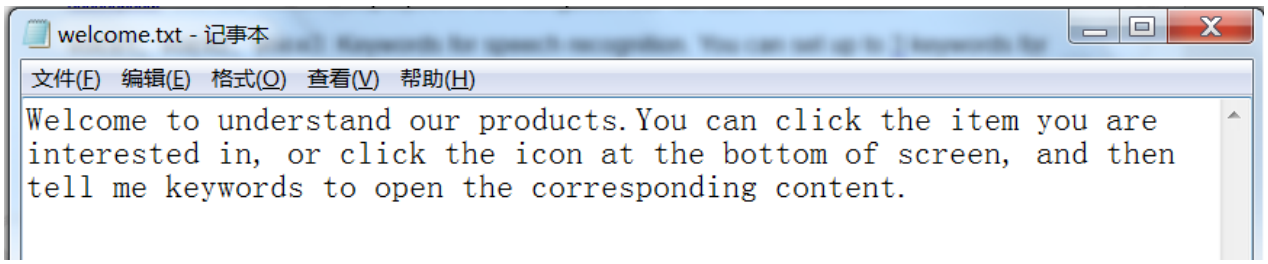
	A	B	C	D	E	F	G
1	filename	number1	number2	category	voice1	voice2	voice3
2	Elf for Reception	1	1	Elf Robot	reception	welcome	service
3	Elf for multi-scene	1	2	Elf Robot	multi-scene	application	scene
4	Sanbot Max (audio)	2	3	Sanbot Max	Max	Sanbot Max	audio
5	Sanbot Max Introduction	2	4	Sanbot Max	Introduction		
6	Mini ELF Introduction	3	5	Mini ELF	Mini ELF	Mini	
7	Mini ELF for TEMP	3	6	Mini ELF	temperature	measurement	temp

⑥ Open "welcome.txt" and write the welcome message. When a user opens "Product Introduction" APP, the robot will say the specified content. For example:



Step 2: Copy the "products" folder to the **root directory** of U disk, and insert the U disk into the USB port of robot.

Tips: Remove the fixing piece on the battery compartment on the back of the robot, use a screwdriver to take out the screw on the battery cover, and remove the battery cover to see the USB port.

Step 3: Open "Product Introduction" APP, wait for the "Resource folder has been found in USB stick" prompt on the interface, and click "CONFIRM" to start uploading. Do not remove U disk or close the app during uploading.

Step 4: After the prompt disappears, the content of product introduction has been imported into the robot successfully.

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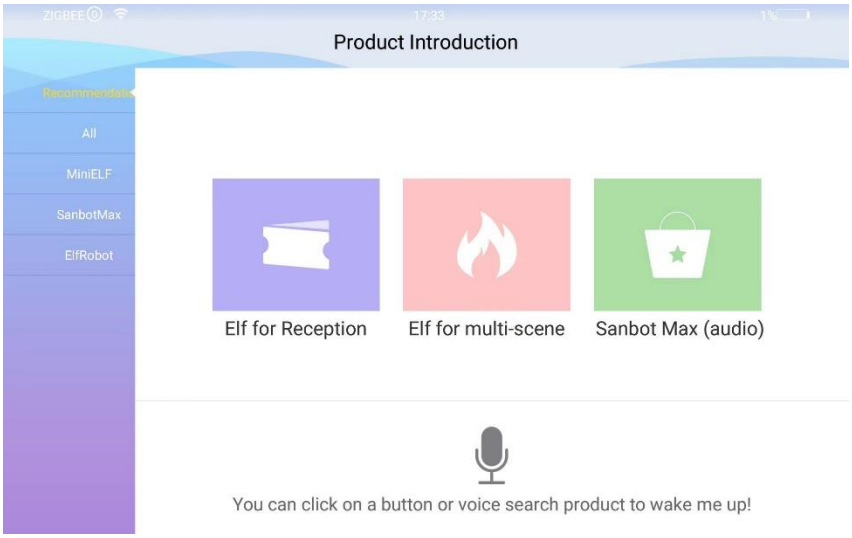

2.6.24.2 Begin to Introduce

After importing the content of product introduction into robot, open "Product Introduction" APP, you have two ways to search for products:

Method 1: Clicking. Click a category in the navigation bar on the left to view the introduction files, and click a file you are interested in to open it.

Method 2: Voice. When the robot is in the awake state, say the product keywords to robot, the robot will automatically search for products matching the keywords, and open the corresponding introduction file.

When playing a video or an audio, support play at the specified location, pause / play, etc. When playing a PPT, the page switches every 10 seconds automatically and says the remarks of PPT. When displaying a picture, it will exit automatically after 10 seconds.

Quit playback: click the hover button ; or click "Quit" in the lower right corner; or wait for the playback to end.

2.6.25 Desktop Replace

Change the apps and background on the desktop. The operation method is as follows:

1) If you need to change the default desktop background, copy the image in the U disk to robot via “File Manager” APP. (The best resolution of image is 1920*1200. Support jpg, jpeg and png format. See chapter "2.6.5 File Manager" for copying the file.
2) Go to the menu and open the “Desktop Replace” APP.
3) If you need to change the desktop background, click "+" to select the image you have copied to the robot.
4) You can put the frequently used apps on the desktop of robot. Select the apps that need to display on the desktop (up to display 4 apps).
5) Click "Preview" to view the modified desktop effect in advance, and click the hover button to return; Click "SAVE".

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Tips:

① Some functions support voice control commands, see chapter "2.7 Voice Control" for details.

② You can stop or pause some functions by touching the robot's touch-sensitive area or saying "Hello Sanbot" to wake up the robot. Touching the top of head can exit music, movie, dance, wander, etc.

③ More and more apps are still coming. You can open the "APP Market" to download the APP.

2.7 Voice Control

2.7.1 Wake-on-voice

The ear light goes out when the robot is in sleep state. Say "Hi, Sanbot" or "Hello Sanbot" to waken it and the ear light turns green. Now you can control the robot by voice command (see table) after you configured your own voice project on dialogflow platform (reference "Google dialogflow build instruction") and chat with it if it connects to the network.

Notes: You are able to waken the robot within 4 meters around it. But it may be affected by the ambient noise. The best way to waken the robot is that you directly face the robot in a distance about 1.5 meters. If no voice is detected within about 10~30 seconds, the robot will get into sleep.

2.7.2 Voice Command

The robot can directly respond to those commands with wake-up words, for example, "Hi, Sanbot, play music" or "Hello Sanbot, play music".

Only in the wake-up state, the robot can respond to those commands without wake-up words, for example, "play music".

2.7.3 Control Command

Definition	Intent Name
control robot forward	Sanbot_Goforward
control robot turn right	Sanbot_Turnright
control robot turn left	Sanbot_Turnleft
control robot turn about	Sanbot_Turnabout
control robot stop	Sanbot_Stop
control robot come to you	Sanbot_Comehere
control robot leave you	Sanbot_Getout
close robot voice	Sanbot_Silent
control robot start to wander	Sanbot_Starttowander
control robot stop to wander	Sanbot_Stopwandering
control robot open projector	Sanbot_Openprojector
control robot close projector	Sanbot_Closeprojector
control robot turn on light	Sanbot_Lighton
control robot turn off light	Sanbot_Lightoff
control robot play music	Sanbot_Playmusic

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Definition	Intent Name
control robot pause music	Sanbot_Pausemusic
control robot resume music	Sanbot_Resumemusic
control robot close music	Sanbot_Closemusic
control robot play previous music	Sanbot_Previousmusic
control robot play next music	Sanbot_Nextmusic
control robot play movie	Sanbot_Playmovie
control robot pause movie	Sanbot_Pausemovie
control robot resume movie	Sanbot_Resumemovie
control robot close movie	Sanbot_Closemovie
control robot play previous movie	Sanbot_Previousmovie
control robot play next movie	Sanbot_Nextmovie
control robot start dance	Sanbot_Startdancing
control robot pause dance	Sanbot_Pausedancing
control robot resume dance	Sanbot_Resumedancing
control robot stop dancing	Sanbot_Stopdancing
control robot go to charge	Sanbot_Charger
control robot increase volume	Sanbot_Volumeup
control robot decrease volume	Sanbot_Volumedown
control robot open specified app	Sanbot_Openapp
control robot quit specified app	Sanbot_Closeapp
control robot take photo	Sanbot_Photo

3. Operate the "Q-Link" APP

3.1 Download and Install

Scan the following QR code on your mobile phone to download and install the "Q-Link" APP.

For Android

For iOS

APP Icon

Tips:

1. "Q-Link" APP is a robot communication and control software. If there is no response after scanning the QR code, please open it from browser.
2. If you have purchased the multilingual version (x.x.x.118) of the robot, please download and install the “Q-Link” APP. If you have purchased the service version (x.x.x.161) of the robot, please

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download and install the “SanLink” APP. If you cannot find these APPs, please enter the "Download Center" on the official website (http://www.sanbotcloud.com/) to download.

3. The accounts of "Q-Link" APP and "SanLink" APP are common, and there is no need to register again.
4. It is recommended that you use a mobile phone with Android 4.0 + or iOS 9 +.

3.2 Register and Login

Open the "Q-Link" APP to enter the login interface.

◆ **If you have not registered an account of "Q-Link":**

Click "New user register", input a username, click "Next", input your phone number, click "Auth Code" and input the correct verification code, fill in the password, and click "Register".

Tips: The account and password are 6~18 letters or letters and numbers.

◆ **If you have registered an account of "Q-Link":**

Please input the correct account or phone number and password, click "Login".

If you forget your login password, click "Forgot Password", input your registered phone number, click "Auth Code" and input the correct verification code, set your new password, click "Confirm".

3.3 Operation of “Chats”

You can switch between “Message” and “Contacts” in the "Chats" interface.

In the “Message” interface, you can add friends, add robots, initiate group chats, receive verification messages, receive alarm information, contact robots, contact friends, view historical messages, etc.

3.3.1 Add Friends

Support adding your friends, and enjoy the convenient communication with work partners, family and friends at any time.

Method 1: Click "+" button in the upper right corner of "Chats" page, select "Scan QR Code", and scan your friend's QR code.

Tips: Your friend can open "Q-Link" APP → select "Me" → click head portrait → select "QR Code" to check their QR code; or click "+" button in the upper right corner of "Chats" page → select "Add Contacts" → click QR Code icon.

Method 2: Click "+" button in the upper right corner of "Chats" page, select "Add Contacts", enter your friend's phone number or account, search and add him / her.

3.3.2 Add Robots

Click "+" button in the upper right corner of "Chats" page, select "Scan QR Code", and scan the QR code of robot (Open "Settings" APP of the robot → About → QR code).

The first person who scans the robot's QR code and click "confirm" on the robot becomes its admin.

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If the robot already has an admin, after scanning its QR code, your application about adding this robot will be sent to the "Q-Link" APP of the admin's phone, and the admin will decide whether to allow you to add.

3.3.3 Initiate Group Chats

Click "+" button in the upper right corner of "Chats" page, select "New Chat", select the contact you want to chat with and click "Confirm" to enter the group chat interface.

Group chat: Support sending text, voice, expression, picture, video, and file.

Group settings: Click the button in the upper right corner of group chat interface to enter the group details interface.

Button or Item	Function
⊕	Invite other contacts to join this group chat
⊖	Remove a group member from this group chat (only visible to the group manager)
Group Name	Edit this group's name, click "Finish" to save
Group QR Code	Display this group's QR code. Non-contact friends can scan this QR code to join this group chat. Click "More" to save the QR code picture (valid within seven days from the generation date) to the phone album
Group Announcement	Set this group's notification or important message, click "Finish" to save(only visible to the group manager)
My Alias in group	Edit your nickname in this group
Manage Group	Transfer your manager rights to other group members (only visible to the group manager)
Mute Notifications	After enabling it, the group can receive messages without prompting
Clean Chat	Clear all chat records in this group
Delete and Quit	If you are the group manager, this function is used to dismiss the group If you are not a group manager, this function is used to exit this group chat

3.3.4 Message List

Search: Input the content what you need to search in the search box to find the relevant chatting records.

Group Chats: Select a group to view its historical messages, start a group chat, and change group settings. See chapter "3.3.3 Initiate Group Chats" for details.

Contact with a robot \ friend: Select a robot \ friend to enter the chat interface. Support sending text, voice, emoji, picture, video, video chat, voice chat, file, etc. Click the button in the upper right corner of chat interface to clear chat records, view friend details, set remarks, etc.

Friend Auth message: When someone applies to add your robot or add you as a friend, the message interface will prompt you. You can click to choose whether to agree or not.

3.3.5 Contacts

Switch to "Contacts" in the "Chats" interface. You can view the added robots, friends and groups, and initiate a chat quickly.

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Search: Input the content what you need to search in the search box to find the relevant names of robot and friend.

New Friend: View all added records.

Group Chats: View the added groups, select a group to enter the group chat interface. See chapter "3.3.3 Initiate Group Chats" for details.

Robot \ Friend: View all added robots and friends. Select a robot \ friend to view the account name and nickname, modify the remarks, and send messages or video/voice chats. See chapter "3.3.4 Message List" for details. Long press a friend in the list to delete this friend.

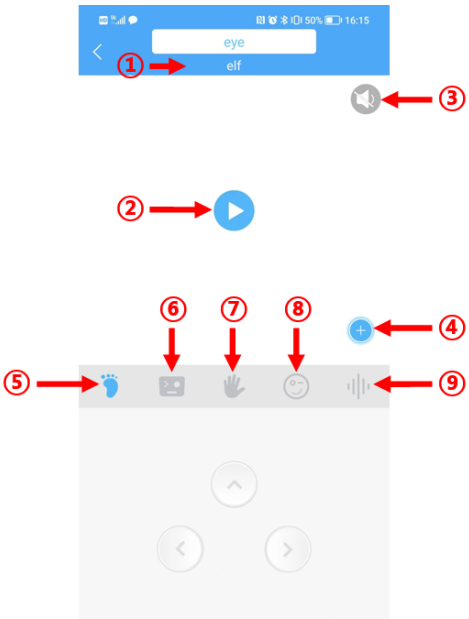
3.4 Operation of “Life”

In the “Life” interface, you can check whether the robot is online (swipe left / right or click button to switch the robot), control the robot remotely, and open or close the robot app remotely.

- Reminders, Trumpet, Safe Home: Configure the corresponding app of robot remotely. The operation method is the same as the operation on the robot screen. Please refer to "2.6 Menu Operation" for details. (The function of Safe Home has been cancelled)
- Basic apps: “Robot’s friends” is used to view the robot's “Contacts” APP, contact the robot's friends, and view the account information of the friends. Video, music, dance and other apps are used to configure the corresponding app of robot remotely. The operation method is the same as the operation on the robot screen. Please refer to "2.6 Menu Operation" for details.
- Robot has installed: View all the installed apps on the robot. Click an app to open this robot app remotely.

3.5 Operation of “eye”

Click button at the bottom to enter the “eye” interface.



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S/N	Function
①	Click the robot name to select other robot you want to control
②	Click this button to play the video of HD Color camera. If the “Open video automatically” function is turned on (See chapter "3.7.5 Settings" for details), it will play the video automatically.
③	While playing the robot’s video, click this button to turn on or off the live sound of robot.
④	Click this button to open the extension buttons: ☞ Click to record a video while playing the robot’s video. Click again to stop recording (recording time needs to exceed 5 seconds). ☞ Click to take a photo for current video while playing the robot’s video. ☞ Click to open or close the LED light of robot remotely. ☞ Click and to adjust the robot’s volume. Click to retract the extension buttons.
⑤	Control the direction of movement for the base plate.
⑥	Control the direction of movement for the head.
⑦	Control the direction of movement for the hand.
⑧	Send eye expression and screensaver (visible when the robot screen is in the screensaver interface) to the robot.
⑨	Click this button to control the robot voice by below methods: 1) Choose a shortcut phrase to let the robot speak this phrase. These shortcut phrases can be modified, added or deleted by clicking "edit". 2) Or click to type a sentence with keyboard, click "send" to let the robot speak this sentence.

Tips:

1. The robot's ear lights turn red when playing the robot video through "Q-Link" APP. Only one user can watch the robot video at the same time. When more than one user watches the robot video, the robot is out of power, the robot is offline, the network is poor, and each of the above could lead to failure watching.
2. When you do not play the robot video, you cannot record a video or take a photo. Photos and videos will save in your phone. You can click "Me" in the "Q-Link" APP, and select "Album" to view.
3. As long as the robot powers on and the network is good, you can control the robot motion regardless of whether the video turn on or not. If you keep pressing the control button, the robot will execute the motion command until it stops when it reaches a limit position. If you release the control button, the robot will stop executing the command.
4. Please confirm that there are no obstacles around the robot when controlling the robot to walk, otherwise the robot cannot move forward or turn around due to the obstacle avoidance. The red \ yellow curve on the video pictures means that the robot is close \ very close to the obstacle.
5. When the robot charges by wire charging, the functions related to movement will fail. See chapter "4.2 Charging Mode" for details.

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3.6 Operation of “Circle”

Display all the robots you have added. Click a robot to enter the device detail interface.

Item	Function
+	Click "+" button in the upper right corner to add a charging pile to the robot. See chapter "4.2 Charging Mode" for details.
remark	Click the edit button next to robot remark to modify the robot remarks
battery level	View the robot's battery level (displayed only when the robot is online)
ID	View the robot's 32-bit ID number
QR Code	Click it to display the robot's QR code. Others can apply to add this robot by scanning this QR code through “Q-Link” APP
Permission management (only visible to the admin)	● **Transfer Identity Of Admin:** Transfer your admin rights to the robot's friends. ● Activate permission: Click “activate” and enter the password when the admin logs into "Q-Link" APP, click “OK” to confirm activation. After that, you need to enter the robot’s login password (default is "admin") when entering the robot menu and "Permission" APP.
Available storage	Display the robot’s remaining storage capacity
Current version	Display the robot’s system version number
Provider	The device name is visible after adding a charging pile to the robot

3.7 Operation of “Me”

3.7.1 Personal Information

Click your head portrait or the area around it at the top of the "Me" interface to enter the personal info interface.

Modify head portrait: Click "Modify Headportrait" (click the picture to enlarge it only), you can select "take picture" to take a picture, or select "choose from album" to choose a picture from phone album and click "OK".

ID: Display your account name.

Alias: Modify your nickname, click "Save".

My phone\Change phone: After you pass the password verification, you can change the phone number bound to this account.

QR code: View your QR code. Others can scan this QR code through “Q-Link” APP to apply for adding you as a friend. See chapter "3.3.1 Add Friends" for details.

Modify password: Input your original password and new password, click "Save".

3.7.2 Album

You can record videos and take photos while playing the robot’s video in the “eye” interface (See chapter “3.5 Operation of eye” for details). Photos and videos will save in the “Album” of your phone.

View files: Click an image file to view it, and click a video file to view the video playback. During the video playback, it supports screenshot, pause/play, turn on/off the video sound, adjust the playback progress, etc.

Delete files: Users of Android phone can long press a file to delete it. Users of iPhone phone can click "Select" to delete it.

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3.7.3 Smart-forum

This function has been disabled.

3.7.4 Tech-support

This function has been disabled. If you encounter a problem, please contact your seller; or send the purchase voucher, robot ID, version number, problem description and picture/video to official customer service (service@sanbot.com).

3.7.5 Settings

- Remind no wifi situation: (Only visible to iPhone phones) If you enable this function, when your phone is not under Wi-Fi network, you enter the “eye” interface and open the video (see chapter "3.5 Operation of eye" for details), it will prompt you to pay attention to the network.
- Open video automatically: If you enable this function, it will play the video of HD Color camera automatically in the “eye” interface (See chapter "3.5 Operation of eye" for details).
- About: View the version number of “Q-Link” APP.
- Binding phone: (Only visible to iPhone phones) Check the phone number bound to this account.
- Safe Exit: Log out safely.

3.7.6 MPS

MPS has not been maintained for a long time, some functions are not available for current version of the robot. So it is not recommended.

Before using MPS, you need to apply for activating your MPS rights. Please contact your dealers or agents. Recommend that you operate the MPS on PC, See “MPS User Manual” for details.

3.7.7 Member Center

It is not suitable for this version of the robot. You do not need to enter.

Tips: The "Q-Link" APP for Android phone is slightly different from iPhone phone. The above introduction is for reference only.

4. About Battery Charging

4.1 Place the Charging Pile

There are two ways to place the charging pile of the robot.

4.1.1 Placing on the Ground

Let the charging metal sheet of the charging pile outward and lean it against a smooth wall. In order not to disturb the robot when it moves, please place the power wire along the wall and **make sure there is no barriers, wires, cords, etc. within two to three meters around the charging pile**. This

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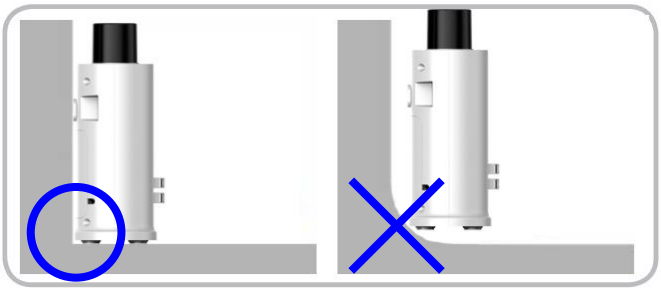
can prevent the robot from tripping or charging failure due to the charging pile's infrared signal reflection by obstacles or obstacle avoidance.



4.1.2 Placing on the Ground and Wall-mounted

Drive two nails into the wall at the appropriate height (the charging pile is still on the ground). Note that the hole spacing is consistent with the charging pile's two wall-mount holes. Lift the charging pile and let the two nails go through the wall-mount holes, and then put the charging pile on the ground so that the charging pile is fixed to avoid from tottering.

4.2 Charging Mode

When you use the charging pile for the first time, please open the cover of the adapter compartments and take the adapter out. Connect the power wire in the packing box with the adapter and then put the adapter back to its original position. Collect wires extending from the left side of the adapter and put them into card slot. Put the power wire into the outlet of the right-hand side and put the cover on the adapter compartments.

Mode one: Auto-charging

(1) Plug the power wire of the charging pile into the power socket.

(2) Add the charging pile to the robot (you can skip this step if a charging pile has been added, and not need to add it again).

① Open the "My Device" APP of robot.

② If you have not added a charging pile, click "Scan QR code to add a device".
If you have added a charging pile, click "Add device".

③ Scan the QR code on the charging pile.

④ Change the device name (such as "charging") and click "Confirm".

⑤ Long press the button on the back of charging pile until the indicator light of charging pile blinks, release the button right now.

Note: If you release the button before or after the indicator light blinks, it will lead to the reset failure.

(If your charging pile uses the old charging strategy, you need to long press the button of charging pile, and release this button when the status of indicator light changes twice. If the indicator light is off, long press the button until the light turns on and then turns off, the light blinks several times to complete the reset. If the indicator light is on, long press the button until the light turns off and then turns on, the light blinks several times to complete the reset.)

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⑥ Refresh the device list (Click "Refresh" button; Press and pull down the device list, then release it to appear a rotating arrow; Re-enter the My Device interface; Restart the robot). The robot has connected to the charging pile when it displays “Connected”.

(3) Open the “Settings” APP of robot → Battery → Enable “Auto recharge” →set the power threshold.

(4) When the battery level is lower than the power threshold, or open the "Charge" APP to enable the charging mode, the robot will actively find the charging pile to charge automatically.

Mode Two: Manual charging

Plug the power wire of the charging pile into the power socket. Move the robot and make the charging metal plates on the back of robot contact with the charging spring strip. Press the button on the charging pile to charge.

Mode Three: Wire charging

Take out the power adapter in the charging pile. Connect one end to the charging plug on the back of robot and the other end to the power socket.

Note: The robot cannot move when it charges by wire charging, so all functions related to movement will be forbidden (such as moving forward, turning around, wandering, dancing, etc.). However, other functions will not be affected (such as playing movie and music, etc.). The robot can move freely when it charges by the charging pile, but it will stop charging if the robot disconnect with the charging pile.

4.3 Battery Indication

- **For the robot (Power on status):**

When the robot is not charging, the wing indicators blink red that means the battery level is less than 40%. The touch screen also displays current battery level.

When the robot is charging, the red wing indicators mean the battery level is less than 40%, the yellow wing indicators mean the battery level is between 40% and 80%, and the green wing indicators mean the battery level is more than 80%. The wing indicators go out after stopping charging.

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◆ **For the charging pile:**

When it is not charging, you can press the emergency charging button to charge it, and the indicator of charging pile keeps yellow.

When it is charging, the indicator of charging pile blinks as BLN. The lower the battery level is, the faster the BLN blinks, and vice versa. The robot will close the charging pile automatically and turn off the indicator after it is fully charged. Long press the emergency charging button for 2 seconds to control the switch of LED indicator.

4.4 Electrical Safety

The charging pile is opened and emits infrared signals only when the robot is looking for it. When the robot finds the charging pile and it is about 50cm away from the robot,

The charging spring strip is electrified with 19V (within safety limits). You and your family members can use securely.

5. Appendix

5.1 Points for Attention

- **Please do not allow preschool children to use this product alone.**
- Juveniles should use this product under the guidance of adults.
- If you allow juveniles to use this product, please be sure to instruct them all do's and don'ts as well as safe use of electricity.
- **Please tell your children not to put their fingers into the gap between the robot's wings and its body as well as the gap between the robot's head and its body to avoid being jammed while the robot moves.**
- **Please softly move the robot head to do the look up and down.**
- Please pay attention to those places where the product may fall or tumble to the ground.
- Please do not immerse the product or the charging pile into any liquid.
- Please use the configured charging pile to charge up the product. Be sure not to use other types of power adapters.
- Please do not throw or kick the product and the charging pile, which may result in mechanical failure, abnormal function as well as degradation of performance.
- Please do not use or store the power adapter in the high temperature, humid or high voltage conditions. Once such abnormal situations as leakage, strange smell, deformation, etc. appear, you should cut the power immediately.
- When you move the robot in the state of power-off, please push its front body forward, i.e. push it go backwards, or push its back slowly at constant speed.

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5.2 FAQ

Q: What are the functions of robot lights on the ears, wings and chassis?

A: If you develop an APP, the robot will light up or blink the color specified by the program when running the APP. If not, see the following table:

<table>
<tr><th>Location</th><th>Number</th><th colspan="2">Robot State</th><th colspan="2">Meaning</th></tr>
<tr><td rowspan="6">Ear</td><td rowspan="6">2 areas</td><td colspan="2" rowspan="5">Non-dancing</td><td>Go out</td><td>The robot is powered off or not woken up.</td></tr>
<tr><td>Green</td><td>The robot is woken up. At this point you can voice control, voice chat, etc.</td></tr>
<tr><td>Red</td><td>Someone watches the robot video through the mobile APP.</td></tr>
<tr><td>Yellow</td><td>Have an unread message. Open the “Message Center” to view.</td></tr>
<tr><td>Blue</td><td>The robot microphone is turned off. The robot will not be awakened by ambient sounds.</td></tr>
<tr><td colspan="2">Dancing</td><td colspan="2">Blink randomly to heighten the dance atmosphere</td></tr>
<tr><td rowspan="5">Wing</td><td rowspan="5">2 areas</td><td rowspan="4">Non-dancing</td><td>Not charging</td><td>Blink red</td><td>Battery level ≤ 40%</td></tr>
<tr><td rowspan="3">Charging</td><td>Red</td><td>Battery level ≤ 40%</td></tr>
<tr><td>Yellow</td><td>40%< Battery level <80%</td></tr>
<tr><td>Green</td><td>Battery level ≥ 80%</td></tr>
<tr><td colspan="2">Dancing</td><td colspan="2">Blink randomly to heighten the dance atmosphere</td></tr>
<tr><td>Chassis</td><td>A circle</td><td colspan="2">Dancing</td><td colspan="2">Blink randomly to heighten the dance atmosphere</td></tr>
</table>

Q: What should I do if the robot cannot find the charging pile?

A: Refer to the following possibilities:

- Make sure that the power wire of charging pile is connected to the power supply.
- Check whether you have not added this charging pile to robot. If so, see chapter "4.2 Charging Mode" for details.
- Check whether the obstacles block the robot or infrared transmitting & receiving area of charging pile. If so, please remove those obstacles.
- Check whether the charging pile is not placed close to the wall, or the surrounding space is too small or narrow.
- Check whether the robot and charging pile are in the different rooms. If so, move the robot to the

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same room where the charging pile is.

- Open the “Hardware Test” APP, check "Infrared Receiver & Emitter Test" and "ZIGBEE Test".

Q: Why can’t the robot charge automatically?

A: Refer to the following possibilities:

- Check whether you disable the robot’s “Auto recharge” function. If so, please access Settings → Battery → Enable “Auto recharge”.
- Check whether the charging pile is not connected to the power supply.
- Check whether you have not added a charging pile for your robot.
- Check whether the battery level of robot has been used up or the robot is powered off.
- Check whether the bottom of charging pile is on the same level with the robot.
- Take out the power adapter of charging pile and charge the robot to check whether the adapter is faulty.

Q: What should I do if the robot has no sound?

A: Check the volume in the settings, turn the volume up; enter the “Hardware detection” APP to select the relevant item; restart the robot; restore factory settings.

Q: What formats does the robot support for video and audio files?

A: You can use common audio and video formats, such as MP3 and MP4.

Supported video formats: MP4, WMV, RMVB, AVI, 3GP, MKV, MPG, VOB, MOV, RM, MPEG, M4V, FLV.

Supported audio formats: MP3, flac, wav, ogg, wma, imy, mid, ac3, mka.

Q: What should I do if I cannot watch the video through "Q-Link" APP?

A: Consider the following possibilities:

- The maximum number of connections exceeded. Only one user can watch the robot video at the same time.
- The network of robot or phone is not connected;
- The robot is turned off or has no power;
- Enter the “Hardware detection” APP to select the relevant item.

Q: How to project the screen content of robot to a large screen (TV, all-in-one PC, PC, etc.)?

A: You can project the robot's screen content to a large screen through a wireless display adapter. Please confirm with the adapter’s customer service that this wireless HDMI display adapter supports Android 4.4. The following operations are for reference only.

Step 1: Connect the wireless display adapter to the power supply and a large screen (TV, all-in-one PC, PC, etc.) as shown below. The large screen will display the operation prompts.

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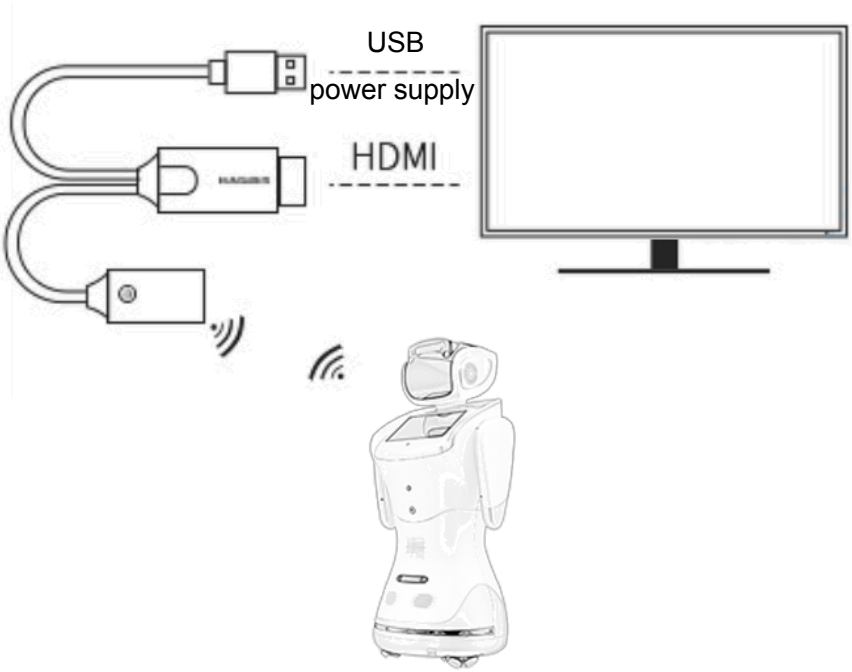
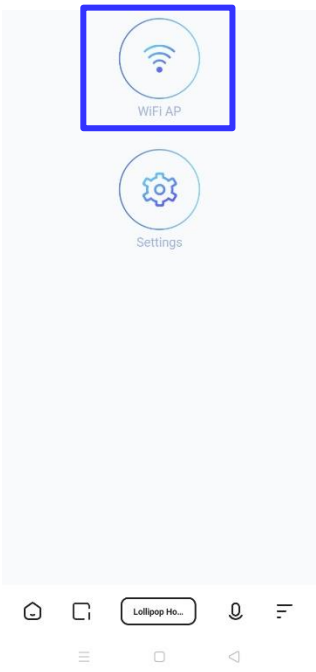


Step 2: Configure the wireless display adapter. When using it for the first time, or using it in a new environment, you need to connect WIFI to the adapter.

① Open the Settings APP of your mobile phone, select the wireless display adapter's name in the WLAN settings and input the password (that is, the device name and password displayed on the large screen) to connect.

② Open the browser of your mobile phone and input the IP address (displayed on the large screen, such as 192.168.49.1).

③ Select "WiFi AP"。



④ Click "Scan" to search for nearby WIFI, select your WIFI and input the password to

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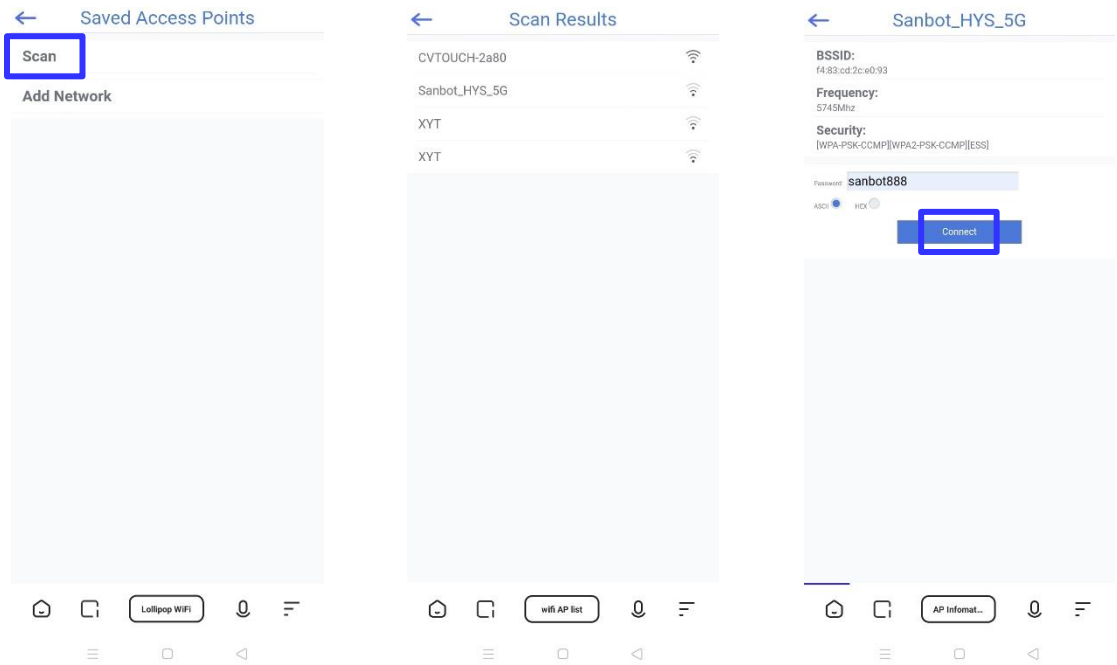
connect.



Step 3: Configure the robot. Open the robot's Settings APP, select your WIFI on the Wi-Fi page and input the password to connect. Then click the adapter's name on the Cast page to project the robot's screen content to a large screen. After enabling "Projected voice", the robot's voice no longer play by the robot, but by the large screen.

Tips:

- ✦ Please use WIFI with better signal to avoid screen stuck or disconnection.
- ✦ The distance between a wireless display adapter and your robot should not be too far (within ten meters). Otherwise, the robot cannot search for it.
- ✦ Please do not move the robot drastically during the projection to avoid screen stuck or disconnection.
- ✦ If the large screen does not have an HDMI interface, please purchase an HDMI connector.
- ✦ For other screen projection problems, please refer to the wireless display adapter's guide, or contact its customer service.

Q: What should I do if I cannot find the wireless display adapter in the "Cast" interface?

A: Consider the following possibilities:

- Check the network connected to the wireless display adapter;
- Check the network connected to the robot;
- Turn off the robot's WIFI, and then turn on the WIFI;
- Restart the robot, wait a few minutes and check again;
- Restart the wireless display adapter, wait a few minutes and check again;
- Restore the factory settings of wireless display adapter.

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5.3 Specifications

5.3.1 Robot Parameters

Item	Spec		
Size	902mm(H) x 421mm(W) x 331mm(D)		
Weight	19kg		
Battery	Li-ion Capacity：20Ah\300Wh Normal use time is about 10h; Stand-by time is more than 20h;		
Sensor	Head	Mic x 7, RGB camera x 1, 3D sensor x 1, Touch sensor x 7	
	Body	PIR sensor x 2, IR obstacle avoidance sensor x 3, IR receiver sensor x 6, Touch sensor x 4, Gyroscope sensor x 1, Electronic compass sensor x1, RGB camera x 1	
	Hand	IR obstacle avoidance sensor x 4, Touch sensor x 2	
	Chassis	IR obstacle avoidance sensor x 10, Omnidirectional wheel x 3	
Moving parts	Head	2 x DOF	180 degree at horizontal direction 30 degree at vertical direction
	Hand	1x2(L\R)DOF	270 degree swing back and forth
	Chassis	3 x DOF	Free angle
Camera	8.0MP HD camera 1.0MP HD color camera with 140 degree aspheric surface angle		
Display	10.1 inch 1080P HD capacitance 10-point touch display		
Projection	Laser 1920 x 720 HD wide gamut output, 60Hz, aspect-ratio: 16:9 (Only apply for the robot with projector)		
Platform	ROS1.1		
Sound	Voice control, 360 degree sound localization 4 x tweeter, 1 x woofer		
Wireless network	WiFi：IEEE 802.11 a\b\g\n(2.4GHz\5GHz)；ZigBee；BT 4.0		
Atmosphere lamp	250 kinds customized colors 2 cycles for ears , 2 pcs for hands, and 1 cycle for chassis		
Moving speed	0.8m/s max		
Climb height	1.5cm max		
Charger mode	Auto charging, Manual charging, Cable charging		

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5.3.2 Charging Pile Parameters

Item	Spec
Size	242.7mm(H) x 248.5mm(W) x 110mm(D)
Weight	1.35kg(including adapter)
Input	AC 100-240V 50\60Hz
Output	DC 19V\4.75A
Indicator light	1 x LED breathing light
Temperature	0℃ ~ +35℃
Operating temperature (in door)	10% ~ 90%
Operating humidity (in door)	0℃ ~ +35℃
Auto charge distance	0-5 meters, 180° signal covering
Installation	Floor or wall-mounted

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